Advanced Coordinate Geometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Distance, midpoint, and slope combined**

Triangle ABC has vertices $A(-2, 1)$, $B(4, 1)$, and $C(1, 7)$. Find the length of the median from C to the midpoint of AB.

Midpoint of AB: $\left(\frac{-2+4}{2}, \frac{1+1}{2}\right) = (1, 1)$. Length of median from $C(1, 7)$ to $(1, 1)$: $\sqrt{(1-1)^2 + (7-1)^2} = \sqrt{0 + 36} = 6$.

2 **[Foundational] • Target: Equations of circles from given conditions**

Find the equation of a circle with a diameter having endpoints $(2, 3)$ and $(8, 11)$.

Center (midpoint): $\left(\frac{2+8}{2}, \frac{3+11}{2}\right) = (5, 7)$. Radius (half the diameter length): diameter $= \sqrt{(8-2)^2 + (11-3)^2} = \sqrt{36 + 64} = \sqrt{100} = 10$, so radius $= 5$. Equation: $(x-5)^2 + (y-7)^2 = 25$.

3 **[Foundational] • Target: Distance from a point to a line**

Find the distance from the point $(3, 4)$ to the line $3x - 4y + 5 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

$d = \frac{|3(3) - 4(4) + 5|}{\sqrt{9 + 16}} = \frac{|9 - 16 + 5|}{5} = \frac{|-2|}{5} = \frac{2}{5}$.

4 **[Foundational] • Target: Locus problems**

Find the equation of the locus of points equidistant from $(2, 0)$ and $(-2, 0)$.

The locus of points equidistant from two fixed points is the perpendicular bisector of the segment joining them. Since both points lie on the x-axis symmetric about the origin, the perpendicular bisector is the vertical line $x = 0$ (the y-axis).

5 **[Foundational] • Target: Rotations and general transformations**

A point $(4, 0)$ is rotated 90° counterclockwise about the origin. Find the coordinates of the image point, using the rotation rule $(x, y) \to (-y, x)$.

Applying the rule: $(4, 0) \to (-0, 4) = (0, 4)$.

6 **[Foundational] • Target: Dilations and scale factor**

A triangle with vertices $(1, 2)$, $(3, 2)$, $(1, 6)$ is dilated by a scale factor of 3 centered at the origin. Find the coordinates of the image vertices.

Multiply each coordinate by 3: $(3, 6)$, $(9, 6)$, $(3, 18)$.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the distance from the point $(3, 4)$ to the line $3x - 4y + 5 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

$d = \frac{|3(3) - 4(4) + 5|}{\sqrt{9 + 16}} = \frac{|9 - 16 + 5|}{5} = \frac{|-2|}{5} = \frac{2}{5}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A tangent from an external point P touches a circle at T. If the distance from P to the circle's centre O is 13 and the radius is 5, find the length of the tangent segment PT.

Step 1: Radius OT is perpendicular to tangent line PT at point of tangency T, forming right triangle ΔOPT with hypotenuse $OP = 13$ and leg $OT = 5$. Step 2: Apply the Pythagorean theorem: $PT = \sqrt{OP^2 - OT^2} = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = \mathbf{12}$

9 [Intermediate] • Target: Finding the equation of a parabola from three points

A parabola $y = ax^2 + bx + c$ passes through $(0, 3)$, $(1, 6)$, and $(-1, 2)$. Find a, b, and c.

From $(0, 3)$: $c = 3$. From $(1, 6)$: $a + b + c = 6$, so $a + b = 3$. From $(-1, 2)$: $a - b + c = 2$, so $a - b = -1$. Adding $a + b = 3$ and $a - b = -1$: $2a = 2$, so $a = 1$, then $b = 2$. Equation: $y = x^2 + 2x + 3$.

10 [Intermediate] • Target: A multi-part triangle-in-the-plane problem

This question has two parts. Triangle PQR has $P(0, 0)$, $Q(6, 0)$, and $R(3, 6)$. (a) Find the area of the triangle using the coordinate formula $\text{Area} = \frac{1}{2}|x_P(y_Q - y_R) + x_Q(y_R - y_P) + x_R(y_P - y_Q)|$. (b) Verify the area using the base-height formula instead.

(a) Area $= \frac{1}{2}|0(0 - 6) + 6(6 - 0) + 3(0 - 0)| = \frac{1}{2}|0 + 36 + 0| = 18$. (b) Using base $PQ = 6$ (along the x-axis) and height equal to R's y-coordinate, 6: Area $= \frac{1}{2}(6)(6) = 18$ ✓, confirming both methods agree.

11 [Intermediate] • Target: Finding the equation of a line given a point and a perpendicular condition

Find the equation of the perpendicular bisector of the segment connecting $(2, 3)$ and $(8, 7)$.

Midpoint: $\left(\frac{2+8}{2}, \frac{3+7}{2}\right) = (5, 5)$. Slope of the segment: $\frac{7-3}{8-2} = \frac{4}{6} = \frac{2}{3}$. Perpendicular slope: $-\frac{3}{2}$. Equation: $y - 5 = -\frac{3}{2}(x - 5)$, so $y = -\frac{3}{2}x + \frac{15}{2} + 5 = -\frac{3}{2}x + \frac{25}{2}$.

12 [Intermediate] • Target: Translating and reflecting equations of curves

The graph of $y = x^2$ is reflected over the x-axis, then shifted 3 units up. Write the equation of the resulting graph.

Reflecting over the x-axis: $y = -x^2$. Shifting up 3: $y = -x^2 + 3$.

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13 [Intermediate] • Target: Finding vertices of a polygon given transformations

A square has vertices $(0, 0)$, $(4, 0)$, $(4, 4)$, $(0, 4)$. Find the coordinates after translating the square 3 units right and 2 units down.

Add $(3, -2)$ to each vertex: $(3, -2)$, $(7, -2)$, $(7, 2)$, $(3, 2)$.

14 [Intermediate] • Target: Finding the intersection of a line and a parabola

Find the points of intersection of $y = x^2 - 2x - 3$ and $y = x - 3$.

Set equal: $x^2 - 2x - 3 = x - 3$, so $x^2 - 3x = 0$, giving $x(x - 3) = 0$, so $x = 0$ or $x = 3$. Corresponding y-values: for $x = 0$, $y = -3$; for $x = 3$, $y = 0$. Intersection points: $(0, -3)$ and $(3, 0)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: A multi-part quadrilateral classification problem • Cross-Lesson: 7.01 Advanced Plane Geometry

This question has three parts. Quadrilateral $WXYZ$ has vertices $W(0, 0)$, $X(6, 0)$, $Y(8, 4)$, $Z(2, 4)$. (a) Find the length of each side. (b) Find the slopes of WX and ZY, and of WZ and XY. (c) Based on parts (a) and (b), classify the quadrilateral as precisely as possible.

(a) $WX = \sqrt{(6-0)^2 + 0^2} = 6$. $XY = \sqrt{(8-6)^2 + 4^2} = \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5}$. $YZ = \sqrt{(2-8)^2 + 0^2} = 6$. $ZW = \sqrt{(0-2)^2 + (-4)^2} = \sqrt{4 + 16} = \sqrt{20} = 2\sqrt{5}$. (b) Slope of WX: $\frac{0-0}{6-0} = 0$. Slope of ZY: $\frac{4-4}{8-2} = 0$ (parallel, since equal slopes). Slope of WZ: $\frac{4-0}{2-0} = 2$. Slope of XY: $\frac{4-0}{8-6} = 2$ (also parallel, equal slopes). (c) Since both pairs of opposite sides are parallel ($WX \parallel ZY$ and $WZ \parallel XY$) and opposite sides are equal in length ($WX = YZ = 6$, $XY = ZW = 2\sqrt{5}$), the quadrilateral is a parallelogram.

16 [Challenging] • Target: Finding the equation of a tangent line to a circle • Cross-Lesson: 8.01 Statistics, Counting & Probability

Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$, using the fact that the tangent is perpendicular to the radius at that point.

The radius to $(3, 4)$ has slope $\frac{4-0}{3-0} = \frac{4}{3}$. The tangent line is perpendicular, so its slope is $-\frac{3}{4}$. Equation: $y - 4 = -\frac{3}{4}(x - 3)$, so $y = -\frac{3}{4}x + \frac{9}{4} + 4 = -\frac{3}{4}x + \frac{25}{4}$.

17 [Challenging] • Target: Finding the vertex and axis of symmetry of a parabola from three points • Cross-Lesson: 1.01 Advanced Number Theory

A parabola passes through $(-1, 0)$, $(3, 0)$, and $(1, -8)$. Find the axis of symmetry and the vertex.

Since the parabola has zeros at $x = -1$ and $x = 3$, the axis of symmetry lies midway between them: $x = \frac{-1+3}{2} = 1$. Since $(1, -8)$ lies exactly on the axis of symmetry $x = 1$, this point IS the vertex: $(1, -8)$.

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18 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Advanced Algebra & Systems

A point $(4, 0)$ is rotated 90° counterclockwise about the origin. Find the coordinates of the image point, using the rotation rule $(x, y) \to (-y, x)$.

Applying the rule: $(4, 0) \to (-0, 4) = (0, 4)$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Advanced Functions

This question has two parts. Triangle PQR has $P(0, 0)$, $Q(6, 0)$, and $R(3, 6)$. (a) Find the area of the triangle using the coordinate formula $\text{Area} = \frac{1}{2}|x_P(y_Q - y_R) + x_Q(y_R - y_P) + x_R(y_P - y_Q)|$. (b) Verify the area using the base-height formula instead.

(a) Area $= \frac{1}{2}|0(0 - 6) + 6(6 - 0) + 3(0 - 0)| = \frac{1}{2}|0 + 36 + 0| = 18$. (b) Using base $PQ = 6$ (along the x-axis) and height equal to R's y-coordinate, 6: Area $= \frac{1}{2}(6)(6) = 18$ ✓, confirming both methods agree.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Advanced Trigonometry

A line passes through $(2, 3)$ and is parallel to $2x - y = 5$. Find its equation in slope-intercept form.

Step 1: Find the slope of $2x - y = 5 \Rightarrow y = 2x - 5$. Slope $m = 2$. Step 2: Parallel lines share identical slopes, so $m_{\text{parallel}} = 2$. Step 3: Use point-slope form with $(2, 3)$:
$y - 3 = 2(x - 2) \Rightarrow y - 3 = 2x - 4 \Rightarrow$ **$y = 2x - 1$**

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the distance from the point $(9, -1)$ to the line $20x - 21y - 11 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{248}{29}$

b. $\frac{219}{29}$

c. -190

d. $\frac{190}{29}$ ✓

$d = \frac{|20(9) - 21(-1) - 11|}{\sqrt{400 + 441}} = \frac{|190|}{29} = \frac{190}{29}$.

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22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the distance from the point $(7, 1)$ to the line $3x + 4y + 9 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{44}{5}$

b. $\frac{39}{5}$

c. $\frac{34}{7}$

d. $\frac{34}{5}$ ✓

$d = \frac{|3(7) + 4(1) + 9|}{\sqrt{9 + 16}} = \frac{|34|}{5} = \frac{34}{5}$.

23 [Foundational] • Target: Distance from a point to a line

Find the slope of the line through $(4, -5)$ and $(8, -2)$.

a. $-\frac{3}{4}$

b. $\frac{3}{4}$ ✓

c. $-\frac{4}{3}$

d. $\frac{4}{3}$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-2 + 5}{8 - 4} = \frac{3}{4}$.

24 [Foundational] • Target: Locus problems

Find the slope of the line through $(-6, -5)$ and $(-7, 6)$.

a. -11 ✓

b. $-\frac{1}{11}$

c. 11

d. $\frac{1}{11}$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{6 + 5}{-7 + 6} = -11$.

25 [Foundational] • Target: Rotations and general transformations

Find the slope of the line through $(-4, 6)$ and $(5, 9)$.

a. -3

b. $\frac{1}{3}$ ✓

c. $-\frac{1}{3}$

d. 3

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{9 - 6}{5 + 4} = \frac{1}{3}$.

26 [Foundational] • Target: Dilations and scale factor

Find the slope of the line through $(-1, -9)$ and $(4, 0)$.

a. $-\frac{9}{5}$

b. $\frac{5}{9}$

c. $\frac{9}{5}$ ✓

d. $-\frac{5}{9}$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0 + 9}{4 + 1} = \frac{9}{5}$.

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27 [Foundational] • Target: Finding the equation of a parabola from three points

Find the slope of the line through $(-5, -1)$ and $(-4, 8)$.

a. -9

b. 9 ✓

c. $\frac{1}{9}$

d. $-\frac{1}{9}$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{8+1}{-4+5} = 9.$

28 [Foundational] • Target: A multi-part triangle-in-the-plane problem

Find the slope of the line through $(-3, 7)$ and $(7, -4)$.

a. $-\frac{11}{10}$ ✓

b. $\frac{10}{11}$

c. $\frac{11}{10}$

d. $-\frac{10}{11}$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-4-7}{7+3} = -\frac{11}{10}.$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Finding the equation of a line given a point and a perpendicular condition

Find the slope of the line through $(-9, -3)$ and $(-7, 1)$.

a. 2 ✓

b. -2

c. $-\frac{1}{2}$

d. $\frac{1}{2}$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1+3}{-7+9} = 2.$

30 [Intermediate] • Target: Translating and reflecting equations of curves

Find the distance from the point $(1, -1)$ to the line $3x - 4y + 6 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{18}{5}$

b. $\frac{13}{5}$ ✓

c. $\frac{23}{5}$

d. -13

$d = \frac{|3(1) - 4(-1) + 6|}{\sqrt{9+16}} = \frac{|13|}{5} = \frac{13}{5}.$

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31 [Intermediate] • Target: Finding vertices of a polygon given transformations

Find the distance from the point $(9, 7)$ to the line $5x + 12y + 11 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{166}{13}$

b. $\frac{140}{17}$

c. $\frac{140}{13}$ ✓

d. $\frac{153}{13}$

$d = \frac{|5(9) + 12(7) + 11|}{\sqrt{25 + 144}} = \frac{|140|}{13} = \frac{140}{13}$.

32 [Intermediate] • Target: Finding the intersection of a line and a parabola

Find the distance from the point $(-2, -2)$ to the line $7x - 24y - 15 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $-\frac{19}{17}$

b. $\frac{44}{25}$

c. $\frac{69}{25}$

d. $\frac{19}{25}$ ✓

$d = \frac{|7(-2) - 24(-2) - 15|}{\sqrt{49 + 576}} = \frac{|19|}{25} = \frac{19}{25}$.

33 [Intermediate] • Target: A multi-part quadrilateral classification problem

Find the distance from the point $(-3, 5)$ to the line $12x + 35y - 15 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{161}{37}$

b. $\frac{124}{47}$

c. $\frac{198}{37}$

d. $\frac{124}{37}$ ✓

$d = \frac{|12(-3) + 35(5) - 15|}{\sqrt{144 + 1225}} = \frac{|124|}{37} = \frac{124}{37}$.

34 [Intermediate] • Target: Finding the equation of a tangent line to a circle

Find the distance from the point $(-5, 5)$ to the line $8x + 15y - 13 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{39}{17}$

b. $\frac{22}{23}$

c. $\frac{22}{17}$ ✓

d. $\frac{56}{17}$

$d = \frac{|8(-5) + 15(5) - 13|}{\sqrt{64 + 225}} = \frac{|22|}{17} = \frac{22}{17}$.

35 [Intermediate] • Target: Finding the vertex and axis of symmetry of a parabola from three points

Find the distance from the point $(2, -3)$ to the line $20x + 21y + 7 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{16}{41}$

b. $-\frac{16}{41}$

c. $-\frac{16}{29}$

d. $\frac{16}{29}$ ✓

$d = \frac{|20(2) + 21(-3) + 7|}{\sqrt{400 + 441}} = \frac{|-16|}{29} = \frac{16}{29}$.

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36 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the y-intercept of the perpendicular bisector of the segment connecting $(-5, -6)$ and $(0, 0)$.

a. $\frac{61}{12}$

b. 0

c. $-\frac{61}{12}$ ✓

d. $-\frac{49}{12}$

Midpoint: $(-5/2, -3)$. Slope of the segment: $6/5$, so the perpendicular slope is $-5/6$. Using $y + 3 = -5/6(x + 5/2)$, the y-intercept is $-\frac{61}{12}$.

37 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the y-intercept of the perpendicular bisector of the segment connecting $(2, -3)$ and $(8, 7)$.

a. -5

b. 5 ✓

c. 6

d. $-\frac{19}{3}$

Midpoint: $(5, 2)$. Slope of the segment: $5/3$, so the perpendicular slope is $-3/5$. Using $y - 2 = -3/5(x - 5)$, the y-intercept is 5.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Advanced Algebra & Systems

Find the sum of the x-coordinates of the intersection points of $y = x^2 - 12$ and $y = -4x$.

a. -12

b. -4 ✓

c. -2

d. -6

Setting the equations equal: $x^2 + 4x - 12 = 0$. By Vieta's formulas, the sum of the roots is $-(4) = -4$.

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Advanced Functions

Find the sum of the x-coordinates of the intersection points of $y = x^2 + 4x - 11$ and $y = -x - 5$.

a. -3

b. -5 ✓

c. -6

d. -7

Setting the equations equal: $x^2 + 5x - 6 = 0$. By Vieta's formulas, the sum of the roots is $-(5) = -5$.

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40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Advanced Trigonometry

A parabola $y = ax^2 + bx + c$ passes through $(0, 5)$, $(1, 8)$, and $(-1, -4)$. Find a.

a. -3 ✓

b. 6

c. -2

d. -4

From $(0, 5)$: $c = 5$. From $(1, 8)$: $a + b + c = 8$, so $a + b = 3$. From $(-1, -4)$: $a - b + c = -4$, so $a - b = -9$. Adding: $2a = -6$, so $a = -3$.

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Pre-Calculus Concepts

A parabola $y = ax^2 + bx + c$ passes through $(0, 7)$, $(1, 0)$, and $(-1, 12)$. Find a.

a. -6

b. -1 ✓

c. 0

d. -2

From $(0, 7)$: $c = 7$. From $(1, 0)$: $a + b + c = 0$, so $a + b = -7$. From $(-1, 12)$: $a - b + c = 12$, so $a - b = 5$. Adding: $2a = -2$, so $a = -1$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Advanced Plane Geometry

Find the distance from the point $(0, 3)$ to the line $8x + 15y - 9 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{70}{17}$

b. $\frac{36}{17}$ ✓

c. $\frac{53}{17}$

d. $\frac{36}{23}$

$d = \frac{|8(0) + 15(3) - 9|}{\sqrt{64 + 225}} = \frac{|36|}{17} = \frac{36}{17}$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 8.01 Statistics, Counting & Probability

Find the distance from the point $(8, 0)$ to the line $12x - 35y - 8 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{162}{37}$

b. $\frac{125}{37}$

c. $\frac{88}{37}$ ✓

d. $-\frac{88}{23}$

$d = \frac{|12(8) - 35(0) - 8|}{\sqrt{144 + 1225}} = \frac{|88|}{37} = \frac{88}{37}$.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Advanced Number Theory

What are the coordinates of the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$?

a. $(\pm 4, 0)$ ✓

b. $(0, \pm 4)$

c. $(\pm 5, 0)$

d. $(0, \pm 3)$

Since $a^2 = 25 > b^2 = 9$, the major axis is horizontal. $c^2 = a^2 - b^2 = 25 - 9 = 16 \Rightarrow c = 4$. Foci are at $(\pm 4, 0)$.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Advanced Algebra & Systems

What is the focal length (distance from center to a focus) of $\frac{x^2}{144} - \frac{y^2}{25} = 1$?

a. 11

b. 12

c. 13 ✓

d. 17

For a hyperbola, $c^2 = a^2 + b^2 = 144 + 25 = 169 \Rightarrow c = 13$.

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